

Module 08: Planning — Computational Psychiatry and Clinical Applications

Course: Active Inference 401 | **Unit:** Neuroscientific Frontiers | **Audience:**
Advanced undergraduates / graduate students

Learning Objectives

1. Analyze how **psychiatric conditions** can be understood as disorders of Active Inference — aberrant precision, model misspecification, and inference failure.
2. Evaluate the **computational psychiatry** approach to specific conditions: schizophrenia, depression, anxiety, autism, and addiction.
3. Examine the implications of Active Inference for **treatment development** and personalized medicine.

Key Concepts

1. Computational Psychiatry: The Framework

Computational psychiatry uses formal computational models to understand mental illness. Active Inference provides a unified framework:

The core insight: Psychiatric symptoms are not random malfunctions — they are the *consequences of inference under a faulty generative model*. If the model's parameters are wrong (wrong priors, wrong precision, wrong likelihood mapping), the resulting perception, action, and learning will be systematically distorted in predictable ways.

Three levels of failure:

1. **Precision failure:** The weighting of prediction errors is wrong — too much or too little confidence in sensory data vs. prior beliefs
2. **Model misspecification:** The generative model has wrong parameters — incorrect beliefs about how the world works

3. **Structure failure:** The model lacks the right variables or hierarchical structure to represent reality

2. Schizophrenia: Aberrant Precision

Schizophrenia is modeled as a disorder of **precision weighting** — specifically, abnormally high precision on sensory prediction errors relative to prior expectations:

Hallucinations: When sensory precision is too high, random sensory noise is treated as meaningful signal. The brain "hears" voices because it assigns excessive confidence to noisy auditory prediction errors, interpreting them as genuine external signals rather than internal noise. The gamma oscillation abnormalities observed in schizophrenia support this — elevated gamma reflects excessive error signaling.

Delusions: When the brain over-weights prediction errors, it is compelled to update its model rapidly and dramatically to "explain" these errors. This produces unstable beliefs that change in response to minor evidence — the "jumping to conclusions" bias. Once a delusional model is formed, it can become self-reinforcing (high prior precision prevents further updating).

Negative symptoms (flat affect, social withdrawal, avolition): May reflect a secondary compensatory mechanism — reducing engagement with the world to reduce the overwhelming prediction errors, or may represent a separate precision deficit in the motivational/dopaminergic system.

Dopamine connection: Hyperdopaminergic signaling in the mesolimbic pathway increases the precision (salience) of sensory prediction errors, explaining why dopamine-blocking antipsychotics reduce positive symptoms. The glutamate/NMDA hypothesis connects to weakened top-down predictions (NMDA-dependent feedback), which would leave prediction errors insufficiently suppressed.

3. Depression: Persistently Pessimistic Priors

Depression is modeled as inference under a generative model with **persistently pessimistic priors** and **low expected precision**:

Negative cognitive bias: The depressed brain has strong priors predicting negative outcomes (C vector biased toward aversive states). Positive events generate large prediction errors but are down-weighted (low precision on positive surprises), preventing model updating. This explains the characteristic cognitive triad (negative views of self, world, and future) described by Beck.

Learned helplessness: When repeated negative experiences overwhelm the model's ability to predict controllable outcomes, the agent learns that its actions have no effect (low expected policy precision). This produces passivity and avolition — the agent stops acting because it expects no policy to reduce free energy.

Interoceptive disruption: Depression involves altered interoceptive inference — the brain's model of the body is miscalibrated, producing persistent feelings of fatigue, heaviness, and physical malaise. The anterior insula shows altered connectivity patterns in depression.

Allostatic overload: Chronic stress produces allostatic overload — the homeostatic setpoints (C vector) shift to accommodate chronic threat, making the "expected" state one of high arousal and distress. Recovery requires resetting these priors.

4. Anxiety: Excessive Uncertainty Estimation

Anxiety is modeled as **overestimation of environmental volatility** — the belief that the world is more uncertain and threatening than it actually is:

Hypervigilance: Anxious individuals assign excessive expected precision to potential threats, maintaining high arousal to detect dangers. This is precision over-allocation to threat-relevant prediction errors.

Intolerance of uncertainty: The anxious generative model assigns high cost to uncertainty itself — not just to negative outcomes, but to the state of *not knowing*. This drives excessive information gathering, rumination, and avoidance of ambiguous situations.

Panic disorder: Panic attacks can be understood as catastrophic interoceptive inference — a normal bodily fluctuation (slightly elevated heart rate) is interpreted as evidence of a life-threatening cardiovascular event. The interoceptive model has excessively tight priors about normal heart rate, producing massive prediction errors from minor deviations.

Norepinephrine connection: The locus coeruleus norepinephrine system modulates expected precision of environmental volatility. Overactive LC-NE signaling produces a chronic state of elevated expected volatility — the brain "expects" the unexpected, maintaining constant vigilance.

5. Autism: Precision Imbalance

Autism spectrum conditions have been modeled as a **precision imbalance** favoring sensory detail over temporal and social prediction:

The HIPPEA hypothesis (High Inflexible Precision of Prediction Errors in Autism — Van de Cruys et al., 2014): Autistic perception involves inflexibly high precision on sensory prediction errors, preventing the normal abstraction and generalization that occurs when errors are down-weighted. This produces:

- Enhanced detail perception (weak central coherence)
- Sensory overwhelm (too many high-precision error signals)
- Insistence on sameness (the model struggles to accommodate variability)
- Social difficulties (social signals are inherently variable and ambiguous — requiring flexible precision to interpret)

Interoceptive differences: Alexithymia (difficulty identifying emotions) is common in autism and may reflect altered interoceptive precision — not enough precision on interoceptive signals to differentiate emotional states clearly.

6. Addiction: Habit Precision Hijacking

Addiction is modeled as the **hijacking of policy precision** by a substance or behavior:

Dopaminergic hijacking: Addictive substances produce supraphysiological dopamine release, dramatically increasing the precision of the drug-seeking policy. Over time, this policy becomes dominant — it receives more precision (confidence) than competing alternatives, even when the drug's actual reward has diminished (tolerance).

Model narrowing: The addicted generative model progressively narrows its policy space — fewer and fewer policies are evaluated as the drug-seeking policy dominates. The result is inflexible behavior focused on obtaining the substance.

Withdrawal as prediction error: Withdrawal symptoms are massive interoceptive prediction errors — the body's adapted state (tolerant to the substance) generates catastrophic prediction errors when the substance is absent. The fastest way to reduce these errors is to take the substance again.

7. Treatment Implications

Active Inference suggests specific treatment strategies:

Condition	Computational Target	Treatment Implication
Schizophrenia	Reduce sensory precision	Antipsychotics (reduce DA → reduce error precision); CBT (strengthen top-down priors)
Depression	Update pessimistic priors; restore policy precision	Behavioral activation (generate positive PEs); antidepressants (restore NE/serotonin precision balance)
Anxiety	Reduce expected volatility; recalibrate interoceptive model	Exposure therapy (experience that threats don't materialize → update priors); interoceptive training
Autism	Flexible precision allocation training	Sensory accommodations; social skills training that provides explicit rules (reducing uncertainty)
Addiction	Restore policy competition	Contingency management (increasing precision of alternative policies); mindfulness (increasing awareness of policy evaluation)

Summary

Computational psychiatry through Active Inference provides a unified framework for understanding mental illness as inference under faulty generative models. Schizophrenia involves aberrant precision, depression involves pessimistic priors, anxiety involves uncertainty overestimation, autism involves precision imbalance, and addiction involves policy precision hijacking. This framework suggests specific, mechanistically informed treatment targets.

Further Reading

- Adams, R. A. et al. (2016). Computational psychiatry: towards a mathematically informed understanding of mental illness. *Journal of Neurology, Neurosurgery & Psychiatry*, 87(1), 53-63.
- Stephan, K. E. et al. (2016). Allostatic self-efficacy: a metacognitive theory of dyshomeostasis-induced fatigue and depression. *Frontiers in Human Neuroscience*, 10, 550.
- Van de Cruys, S. et al. (2014). Precise minds in uncertain worlds: predictive coding in autism. *Psychological Review*, 121(4), 649-675.
- Paulus, M. P. et al. (2019). An active inference approach to interoceptive psychopathology. *Annual Review of Clinical Psychology*, 15, 97-122.